

Inequalities – 6

Probabilistic and combinatorial inequalities

10 сентября 2026 года

Theoretical minimum.

Theorem 1 (the three tail bounds). Markov: $\Pr(X \geq t) \leq \frac{EX}{t}$ for $X \geq 0$. Chebyshev: $\Pr(|X - EX| \geq t) \leq \frac{\text{Var } X}{t^2}$. Chernoff and Hoeffding: for a sum of independent variables in $[0, 1]$ the deviation of $t\sqrt{n}$ has probability at most $2e^{-2t^2}$. Each is Markov applied to a cleverly chosen function of X .

Theorem 2 (the probabilistic method). If $EX \geq c$, then some value of X is at least c ; if $\Pr(\text{bad}) < 1$, then a good object exists. Combined with Cauchy–Bunyakovsky in the form $EX^2 \geq (EX)^2$ this proves most counting inequalities of olympiad type.

Fact 1 (double counting and convexity). Counting one quantity in two ways and applying convexity to the result is the combinatorial twin of Jensen: if a graph on n vertices has m edges, then $\sum_v \binom{d(v)}{2} \geq n \binom{2m/n}{2}$, which is how bounds on the number of paths and of cycles are obtained.

-
1. Prove the three bounds of Theorem 1 and show that each is sharp; then prove the second moment bound $\Pr(X > 0) \geq \frac{(EX)^2}{EX^2}$ for a non-negative X .
 2. (IMC 2025 shortlist) Prove that for any two independent identically distributed random variables X and Y taking integer values, $\Pr(|X - Y| \leq 2) \leq 3 \Pr(|X - Y| \leq 1)$.
 3. (IMS 2024) Let $(\Omega, \Pr)$ be a finite probability space and let $E_1, \dots, E_n$ be events such that every $x \in \Omega$ which belongs to E_i belongs to at most a_i of the events. Prove that $\sum_{i=1}^n \frac{\Pr(E_i)}{a_i} \leq 1$.

4. (IMC 2023) Let T be a tree with n vertices and let $d(u, v)$ be the graph distance. Put $W(T) = \sum_{\{u, v\}} d(u, v)$ and $H(T) = \sum_{\{u, v\}} \frac{1}{d(u, v)}$, both sums over unordered pairs of distinct vertices. Prove that $W(T) \cdot H(T) \geq \frac{(n-1)^3(n+2)}{4}$.
5. (Putnam 2018) Determine the greatest possible value of $\sum_{i=1}^{10} \cos(3x_i)$ over reals $x_1, \dots, x_{10}$ with $\sum_{i=1}^{10} \cos(x_i) = 0$.